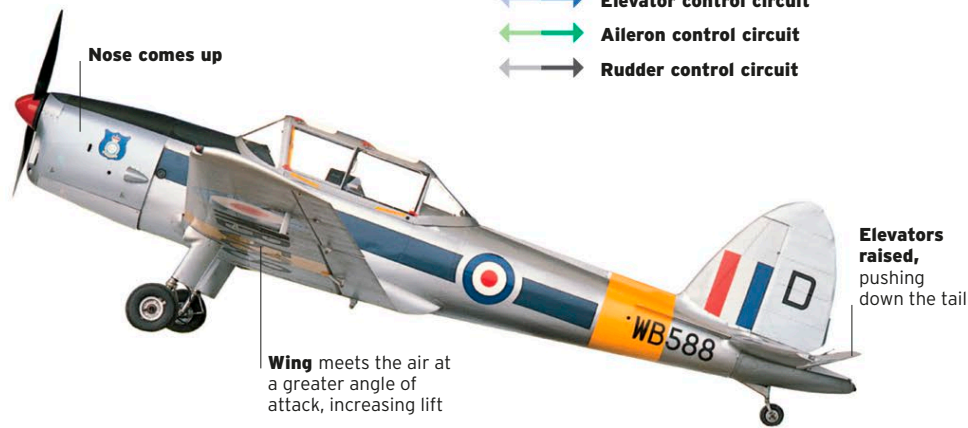
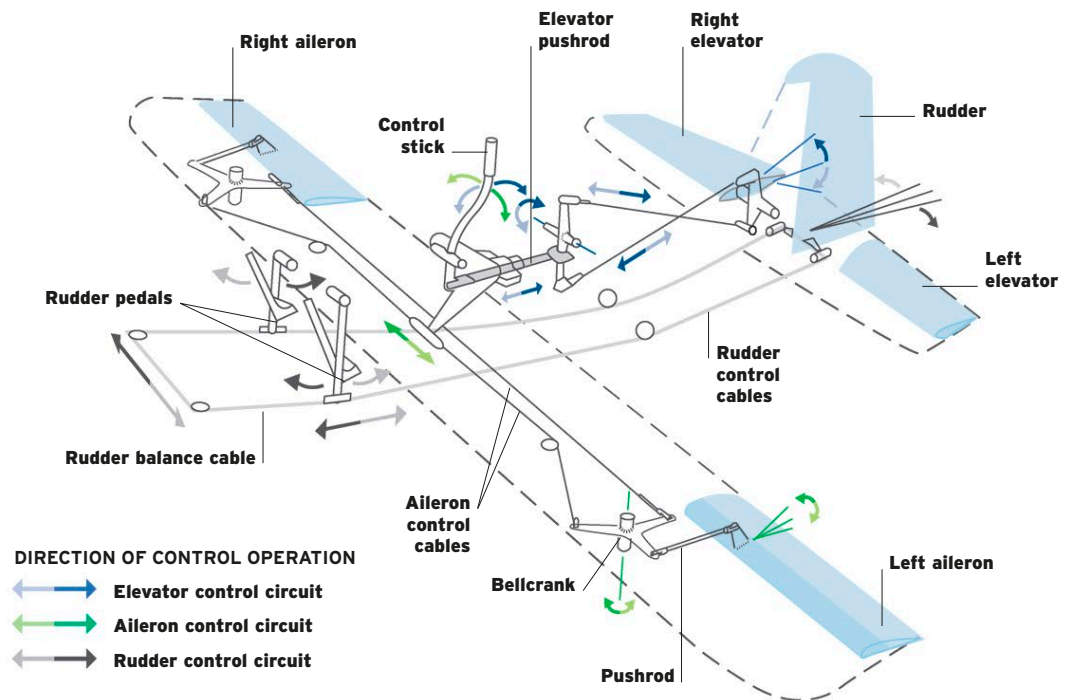


CONTROLLING THE AIRPLANE

A car or a boat is steered left or right, but an airplane has to be controlled in three dimensions. The controls work in three axes: the pilot maintains level flight, or climbs or dives by moving the tail-mounted elevators up and down through pushing the control stick forward and back. The aircraft is kept level, or rolled in one direction or the other, by the ailerons, which are operated by moving the stick sideways. Yaw, or "skidding," is controlled by the vertically mounted rudder, which is operated by foot pedals on the cockpit floor. While large aircraft and fast jets may have powered controls, they are still maneuvered by the same basic stick and rudder input.



PITCHING UP

Pulling the control stick back raises the elevators. The nose rises and the aircraft pitches upward. Since the wing is now operating at a greater angle of attack, giving more lift, the aircraft will start to climb. It will continue to pitch up, unless the stick is returned to neutral.



LEVEL FLIGHT

In level flight the tailplane acts as a stabilizing surface, rather like the feathers of a dart—holding the wing at a steady angle of attack. The elevators will be kept in the neutral position, only being moved up and down by the pilot to compensate for the effects of turbulence.



PITCHING DOWN

Pushing the stick forward lowers the elevators. The nose drops and the aircraft pitches downward. Angle of attack and lift are reduced and the aircraft will start to descend. Because its weight is now acting in the same direction as propeller thrust, speed will build up very quickly.

Left aileron raised, reducing lift from the wing

Right aileron lowered, increasing lift on the wing

ROLLING LEFT AND RIGHT

Moving the stick to one side raises the aileron and reduces the lift on that side, and lowers the aileron and increases the lift on the other, causing the aircraft to roll. As overall wing lift is now acting to the side, the aircraft will start to turn. The rudder is simply used to balance the different amount of drag generated by the up-and-down-going aileron.